



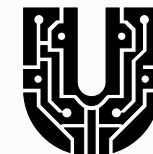
Lit Silicon: A Case Where Thermal Imbalance Couples Concurrent Execution in Multiple GPUs

[Marco Kurzynski](#), [Shaizeen Aga*](#) and [Di Wu](#)

University of Central Florida, AMD*

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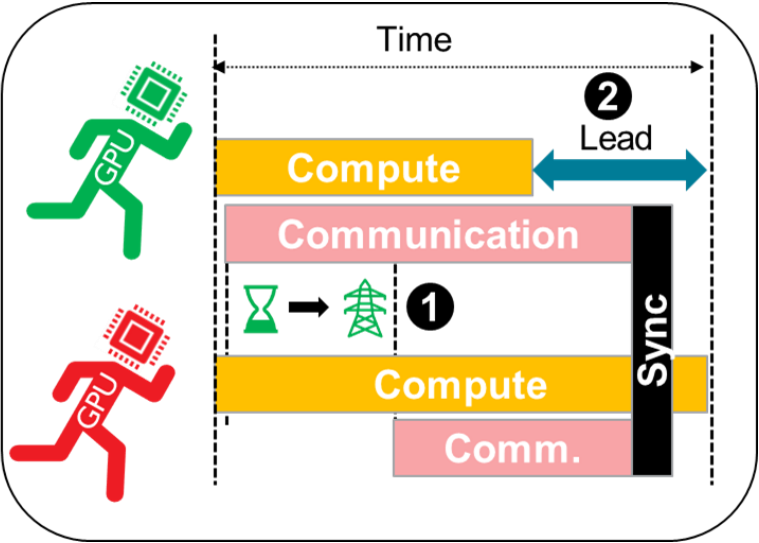
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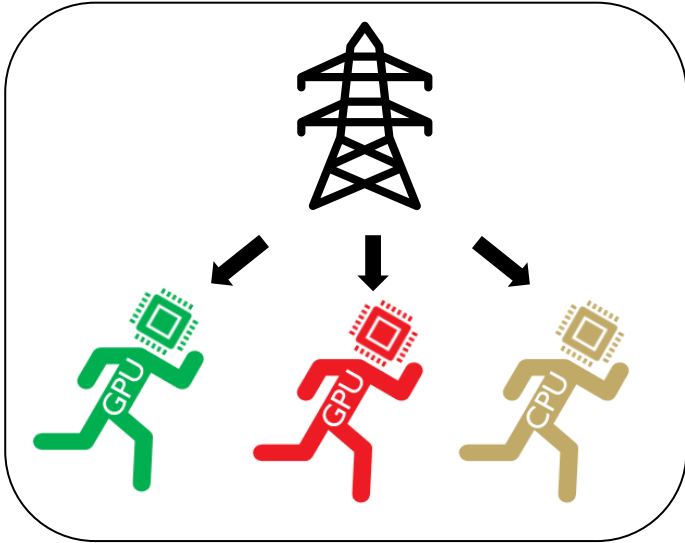
Tackle Lit Silicon → Superior ML Performance And Power

■ Leaders (cooler, faster) ■ Stragglers (hotter, slower)

1 Lit Silicon Effect



2 Variation-aware Power Slicing

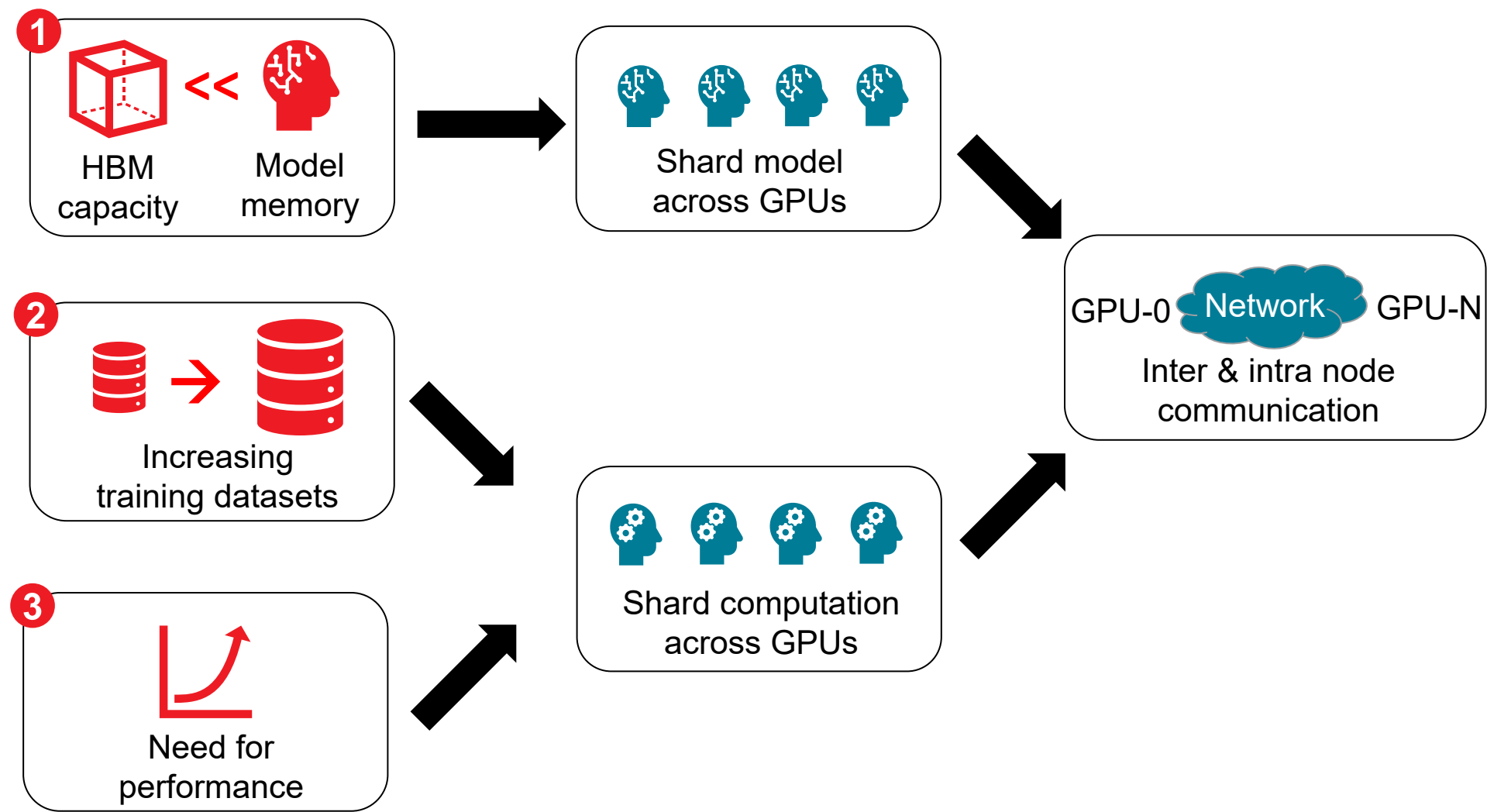


3 Benefits of Tackling Lit Silicon

Tackling Lit Silicon delivers 6% increased throughput

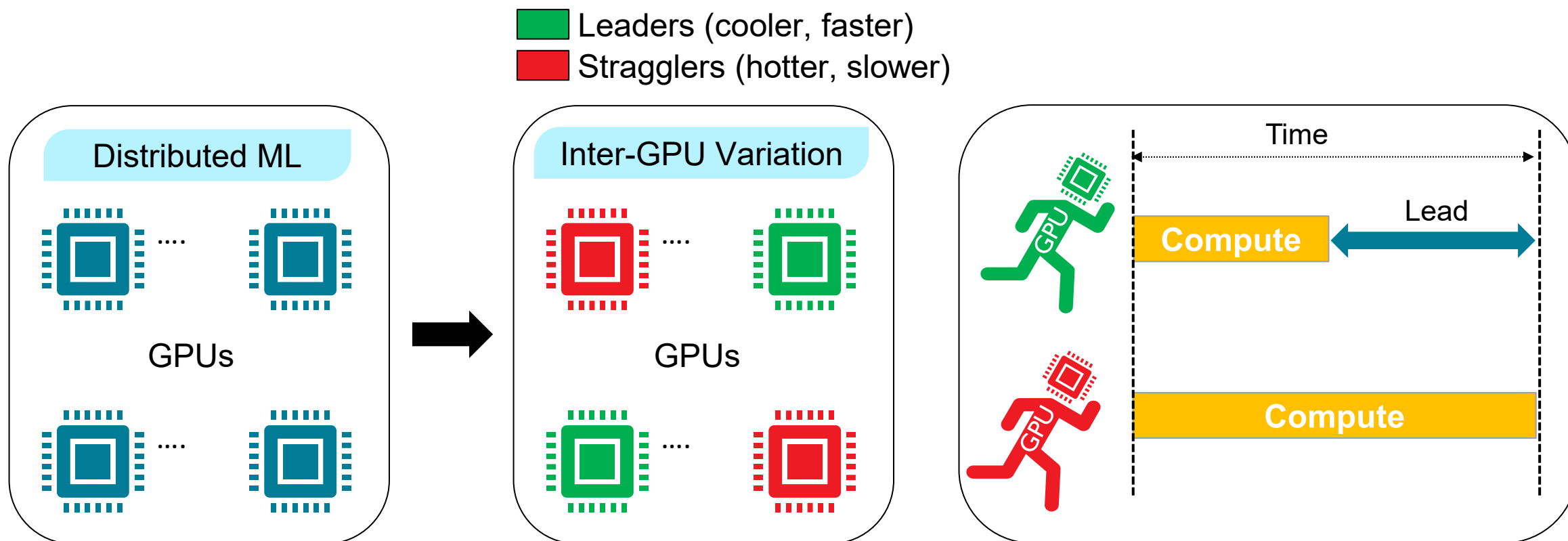
Tackling Lit Silicon delivers 4% power reduction

ML Is Increasingly Distributed → ML Communication



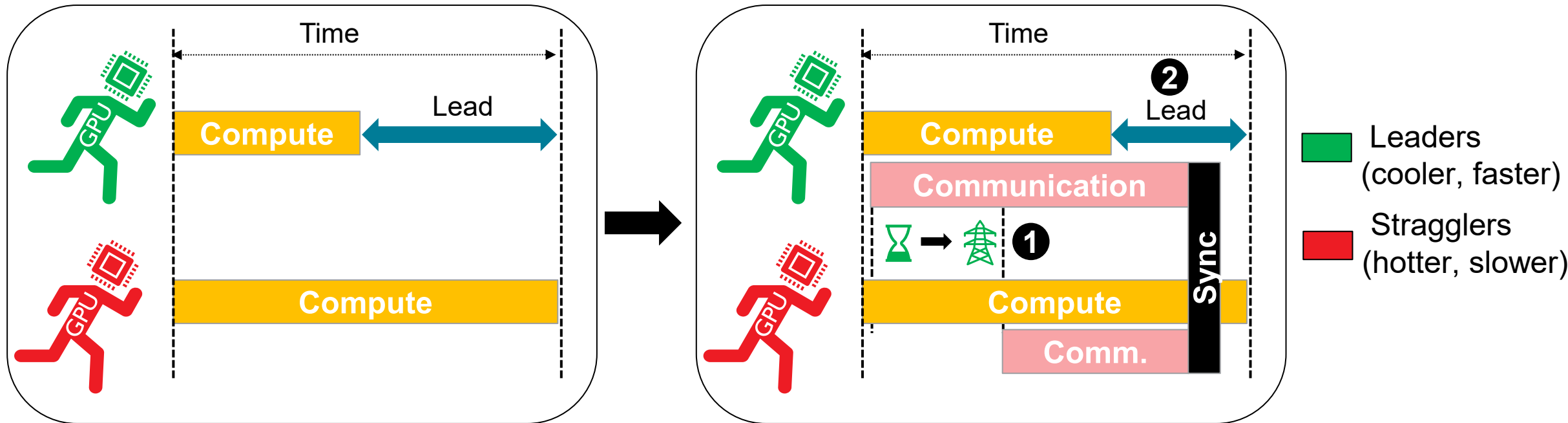
Distributed ML Suffers Thermal Imbalance Induced Variation

- Inter-GPU execution variation = Leader (faster) & straggler (slower) GPUs
 - (e.g., load imbalance, memory allocation differences, etc.)
- **Key observation: thermal imbalance is a key cause for execution variation in distributed ML**



Lit Silicon = Variation x Concurrent Compute-Communication

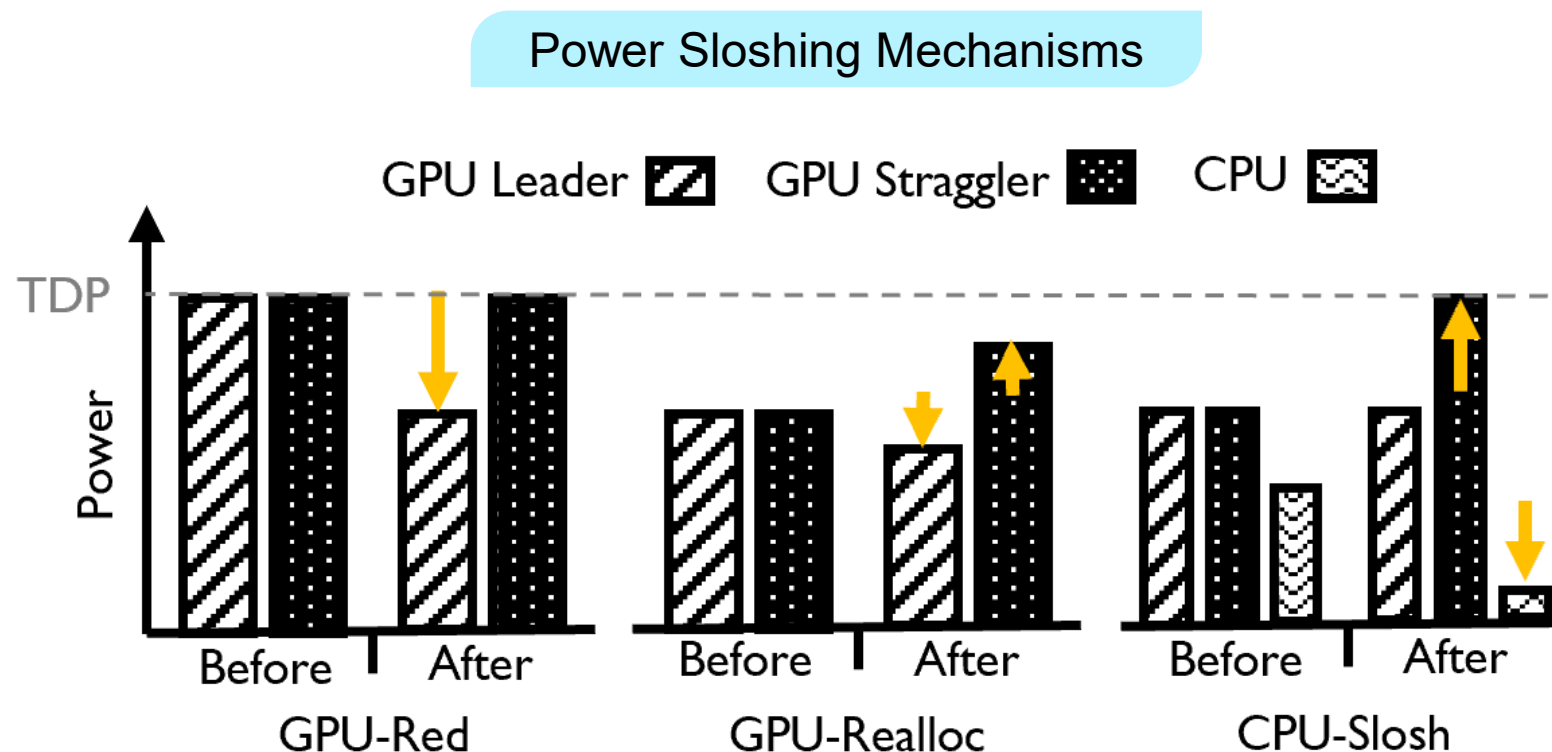
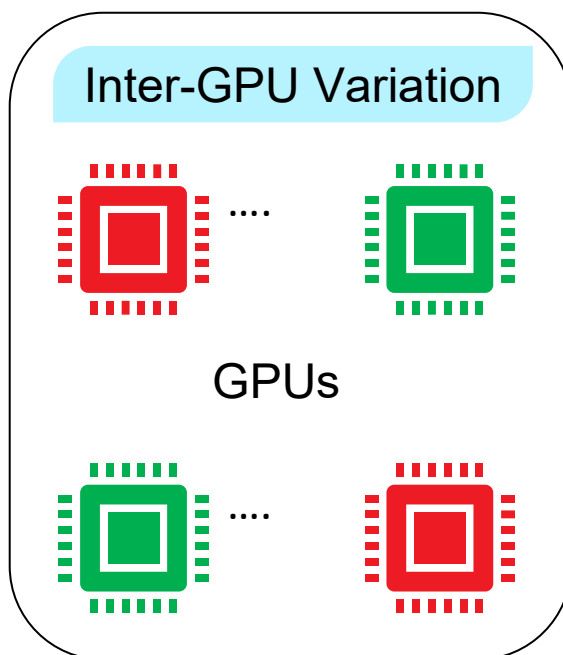
- ML trend: Concurrent compute and communication to hide communication overheads
 - Effect-1: leaders wait for stragglers at sync. points, **extending communication** on leaders
 - Effect-2: **extended communication** on leaders causes **more contention**, slowing them down
- **Key observation:** stragglers extend communication and burn power on leaders, coined the *Lit Silicon effect*



Variation-driven Power Sloshing Can Tackle Lit Silicon

- Execution variation provides key guidance on power re-balancing (sloshing)
 - Leaders can be capped without loss of performance
 - Power can be sloshed to stragglers for superior performance
- **Key observation: detect execution variation & harness it to guide power sloshing decisions**

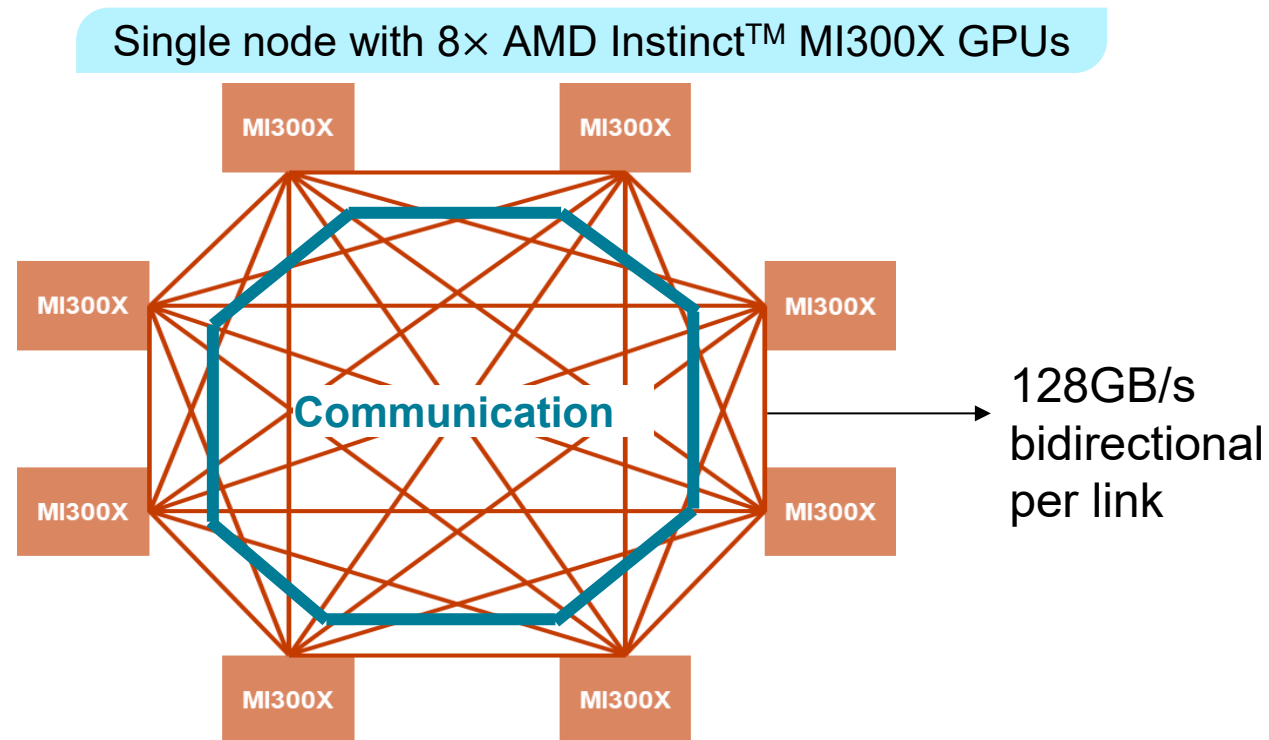
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Experimental Setup

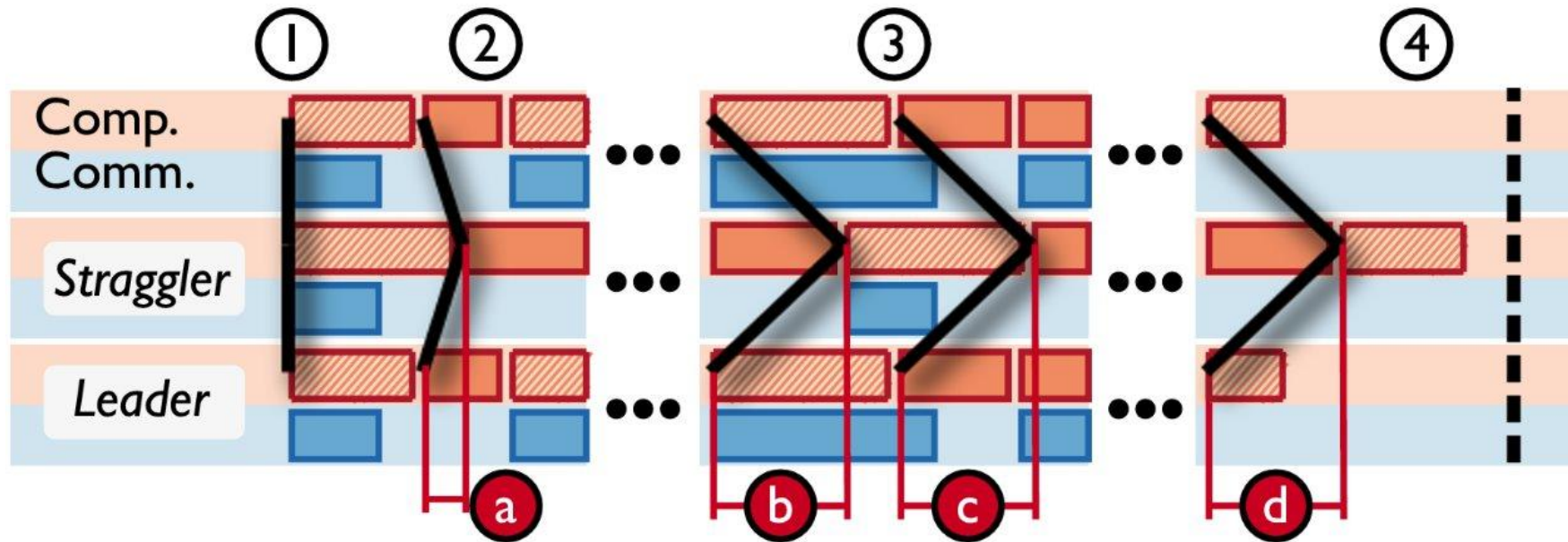
- System evaluated

- **Hardware:** 8× GPU AMD Instinct™ MI300X Platform - fully-connected topology via AMD Infinity Fabric™ technology
- **Software stack:** SemiAnalysis FSDP benchmark (dense), Primus (torchTitan) FSDP benchmark (MoE)*
- **Models:** Llama 3 8B (dense), Mistral 7B v0.1 (dense), Deepseek v3 16B (MoE)
 - MoE with Primus requires round-robin token distribution to experts



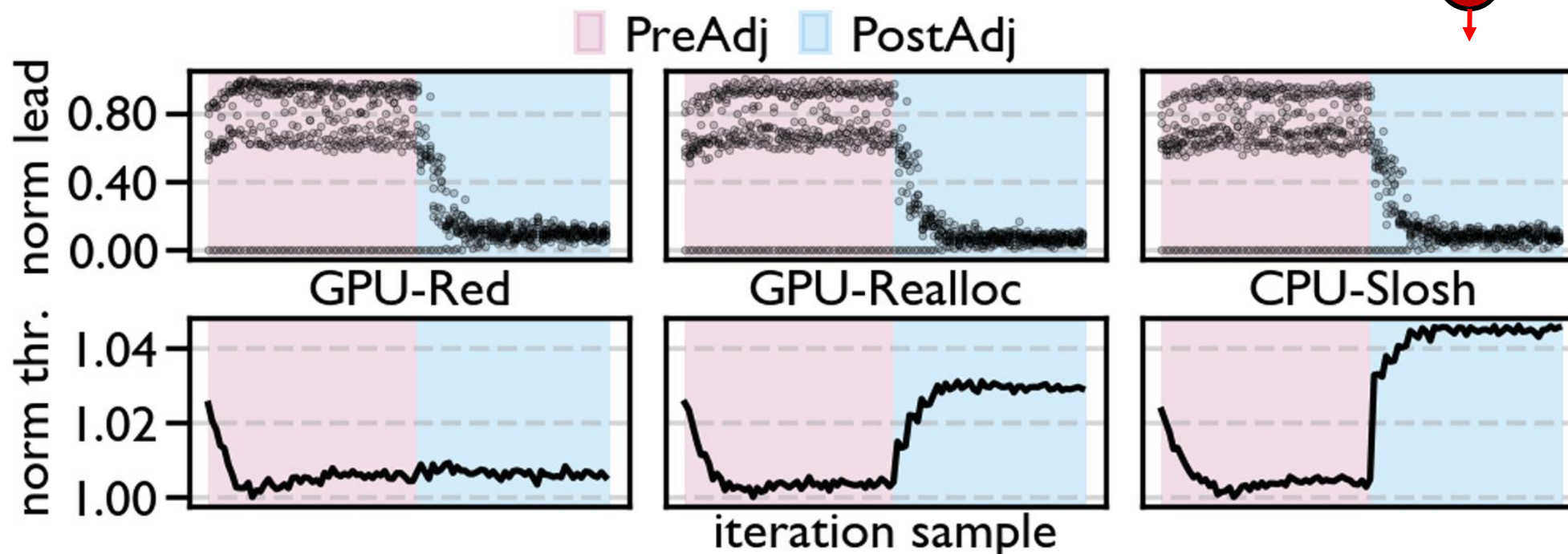
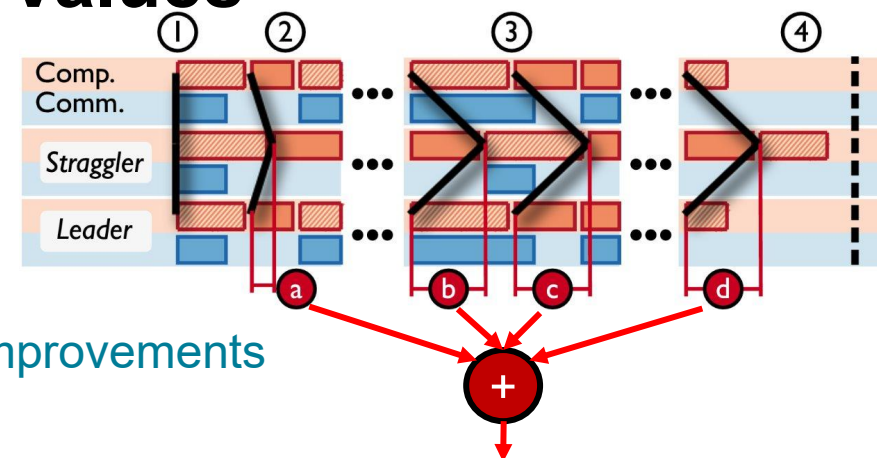
Negative Feedback Loop Of Lit Silicon Over Time

- A “straggler wave” forms by tracking difference in kernel start time (the lead) across GPUs
 1. GPUs start together
 2. Leaders run faster and gain a lead (a)
 3. Leaders suffer from extended communication, wait for stragglers, and reach an equilibrium where lead (b) = (c) = (d)
 4. Leaders wait for the straggler to finish
- Key observation-1: macro execution view (iteration time) fails to convey impact of the Lit Silicon effect
- Key observation-2: fine-grained tracking of "lead" is necessary to tackle the Lit Silicon effect



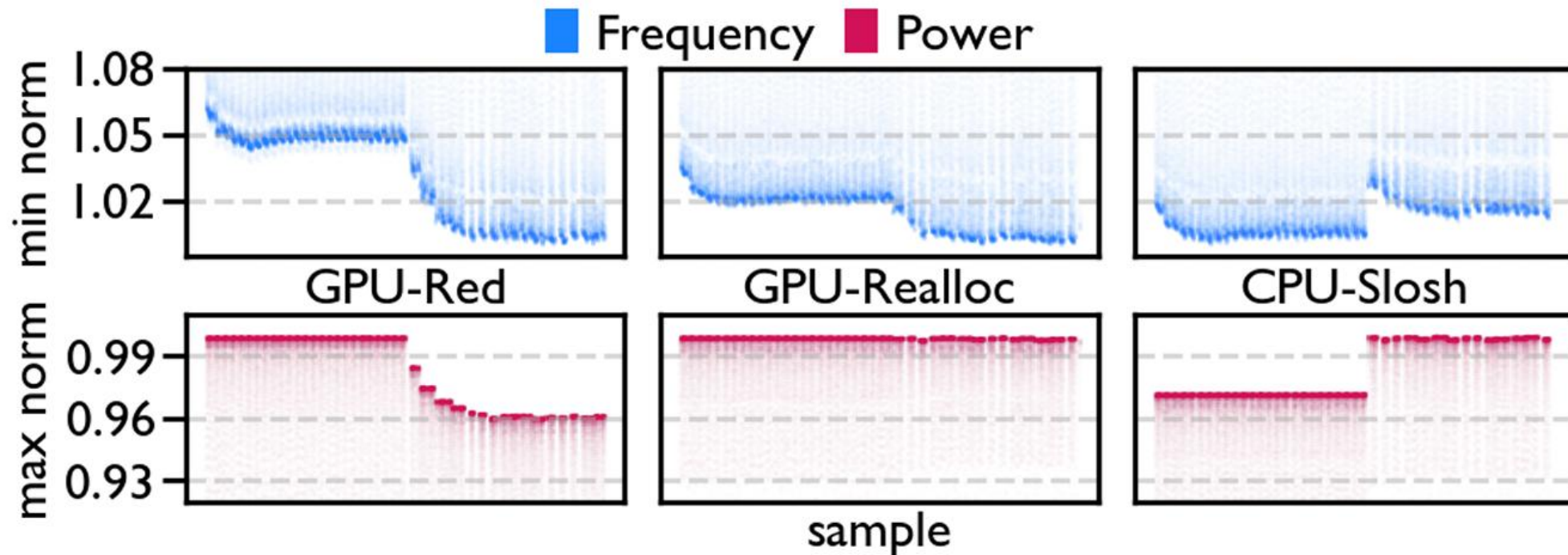
Superior Throughput from Aligning Lead Values

- Impact of power sloshing on lead and throughput
 - GPU-Red maintains throughput ✓
 - GPU-Realloc improves throughput by 3%
 - CPU-Slosh improves throughput by 4%
- Key observation: lead-aware power sloshing delivers throughput improvements



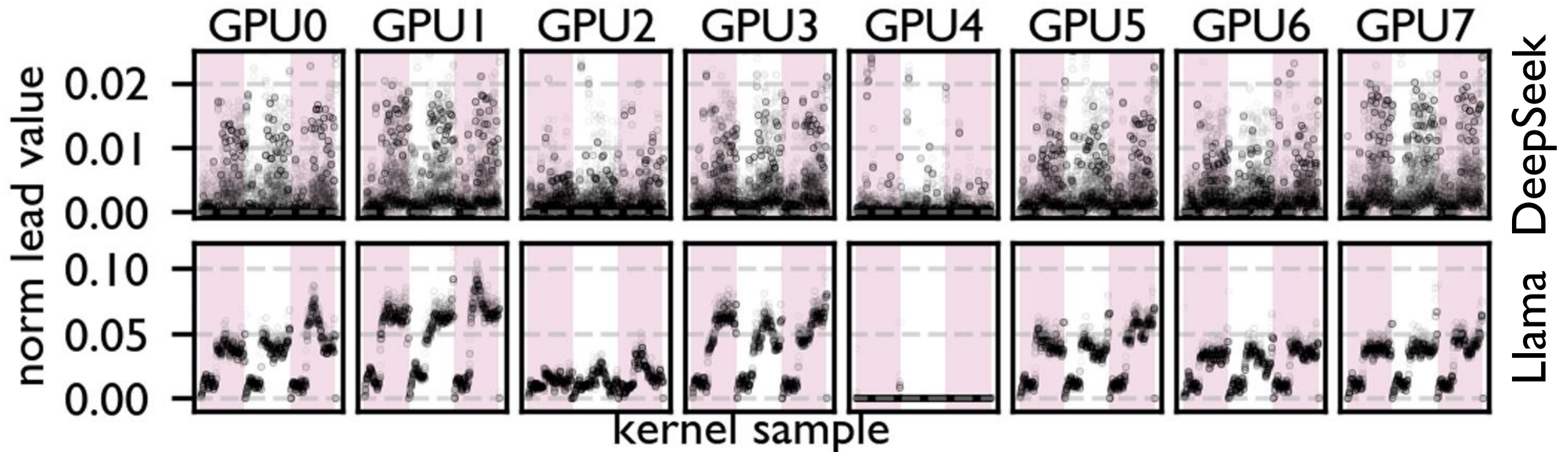
Power Savings from Aligning Lead Values

- Impact of power sloshing on **average GPU** frequency and power
 - GPU-Red saves 4% power
 - GPU-Realloc maintains power ✓
 - CPU-Slosh sloshes 3% more power to GPUs
- Key observation: frequency drops across GPUs, but increases where it matters (straggler)



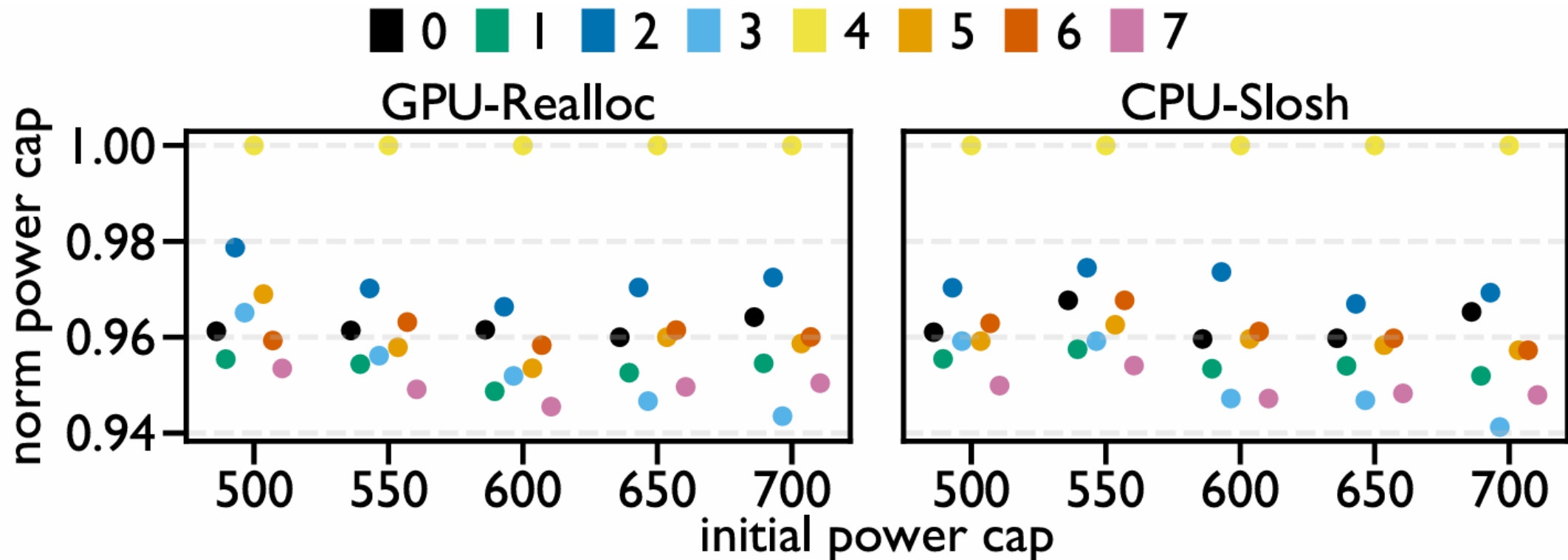
Lit Silicon Prevalent Across ML Models & Algorithms

- Frequent all-to-all communication synchronizes GPUs every layer
 - Smaller lead values with more variation from frequent synchronization makes tuning converge slower
 - Straggler is consistent for MoE and dense workloads
- **Key observation: Lit Silicon manifests in both MoE and dense workloads with the same stragglers & leaders**



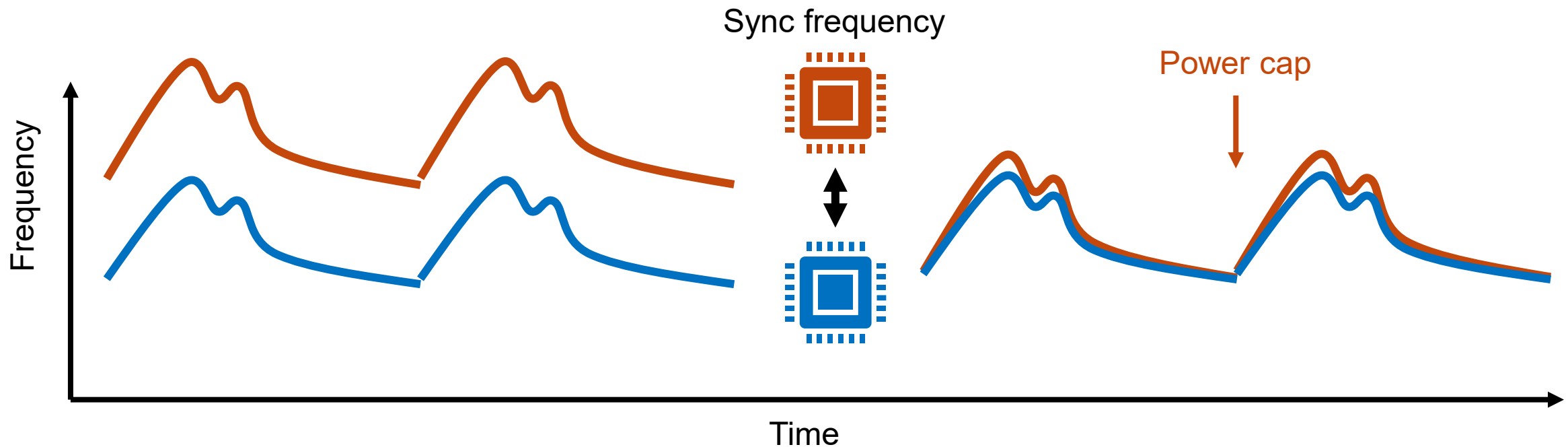
Final Power Caps Are Stable And Reusable

- Node-level power caps can vary, so a power sloshing strategy **must be flexible**
 - The relative power distribution across GPUs is consistent at different uniform node-level power caps
- Key observation: once a power distribution is found, it can be **reused** for any scenario or model (e.g., MoE)



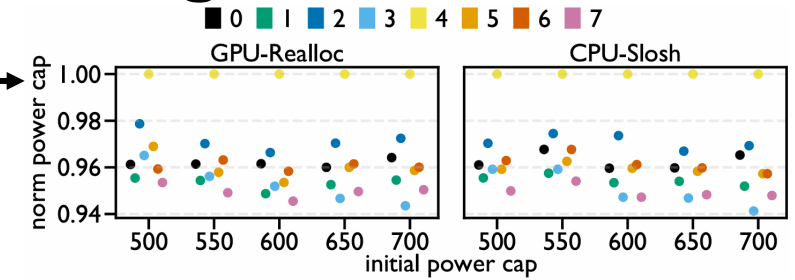
Production-grade Power Sloshing – Online Tuning

- Cons of tuning with lead values:
 - Relies on **administrator privileges** to set power caps
 - Need to dump and process traces for detecting lead values
- **Recommendation:** firmware solution which synchronizes frequencies in the **background**

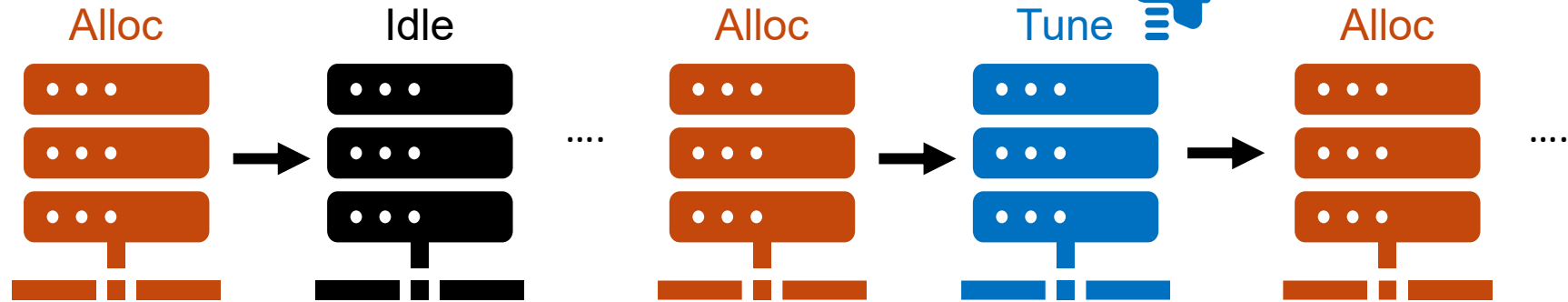


Production-grade Power Sloshing – Offline Tuning

- Curated benchmarks can obtain a power distribution to be re-used
 - **Deployable today**
- Not as efficient or flexible as online tuning
- Our contribution: a curated benchmark for detecting Lit Silicon
- Recommendation: use our curated benchmark to periodically obtain **reusable** power distributions



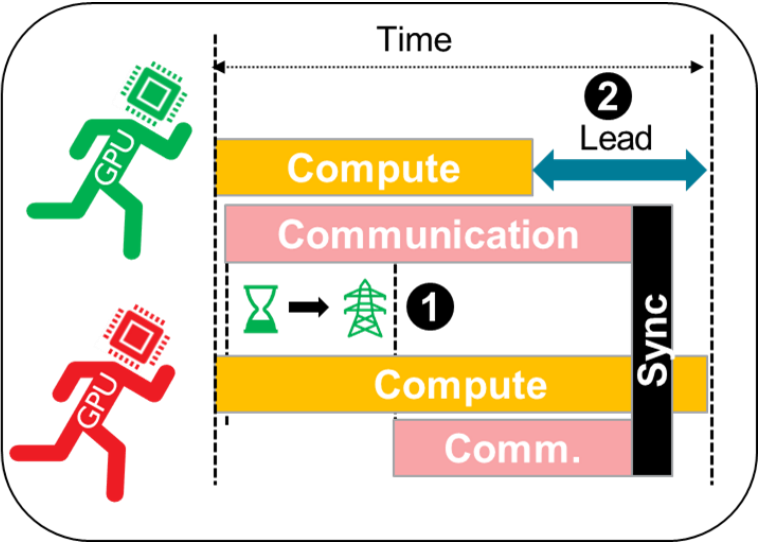
Admin-curated
benchmark



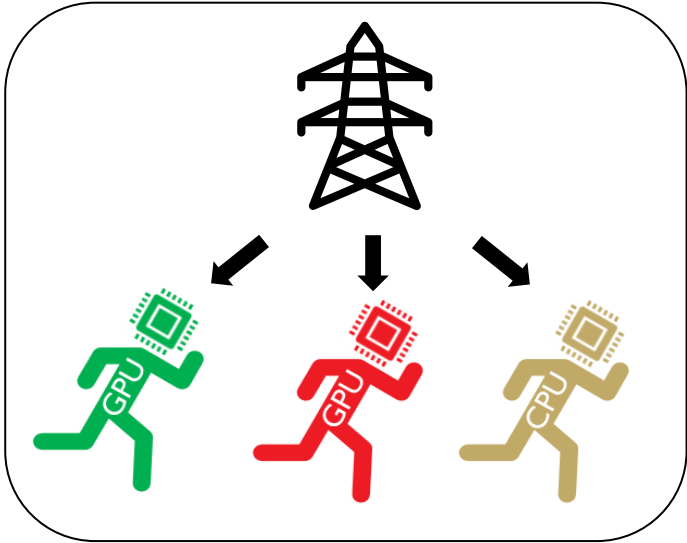
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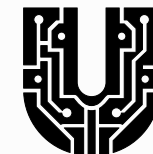
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